

Algorithms 2

Matching in graphs

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1. Matchings in Graphs

- The matching problem is something that students face from the first time that they encounter graph theory.

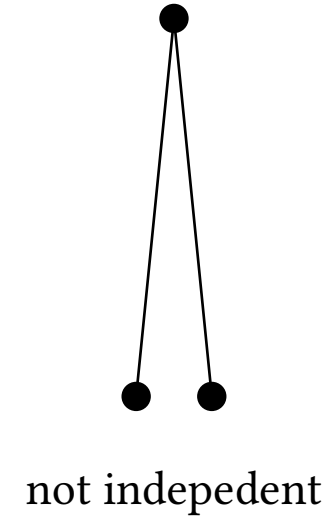
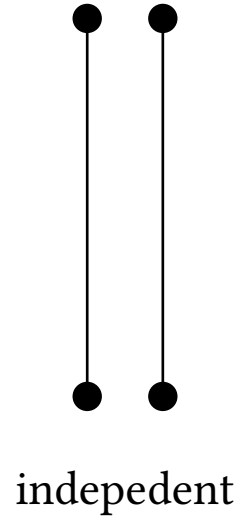
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- Suppose that you have a group of n people, that you want to divide into pairs, how would you go about it? Is it possible that every person finds a pair?

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- What if some people don't want to be paired together, can you do it then?

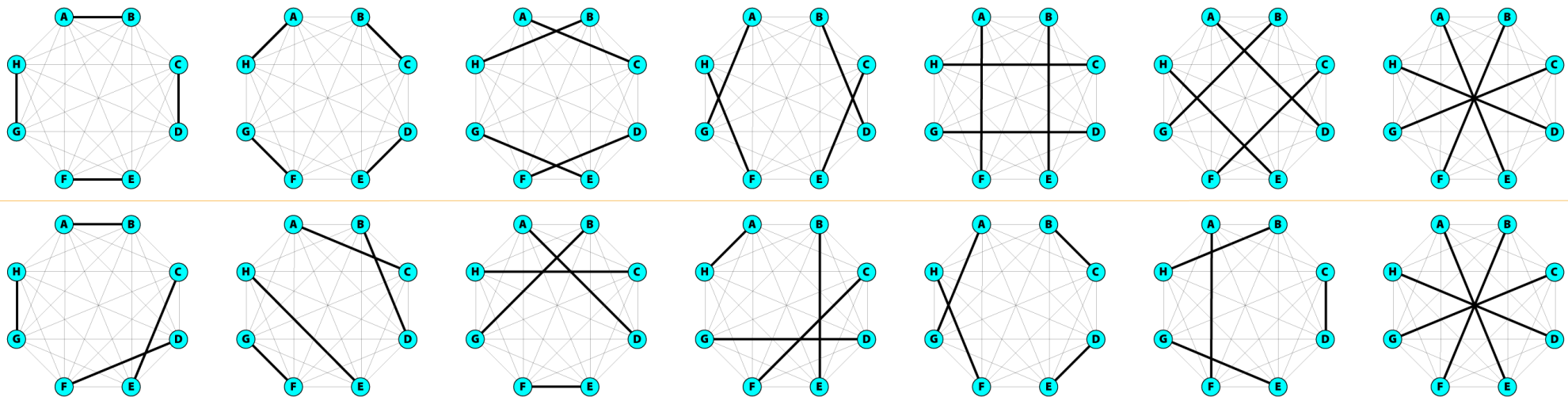
- The matching problem is something that students face from the first time that they encounter graph theory.
- Suppose that you have a group of n people, that you want to divide into pairs, how would you go about it? Is it possible that every person finds a pair?
- What if some people don't want to be paired together, can you do it then?

A common approach is to model this problem with a graph G . Each vertex represents a person, and an edge between two vertices indicates that those two people can be paired together.

For a graph G , two edges $e_1, e_2 \subseteq E(G)$ are called *independent* if there is no common vertex between them.

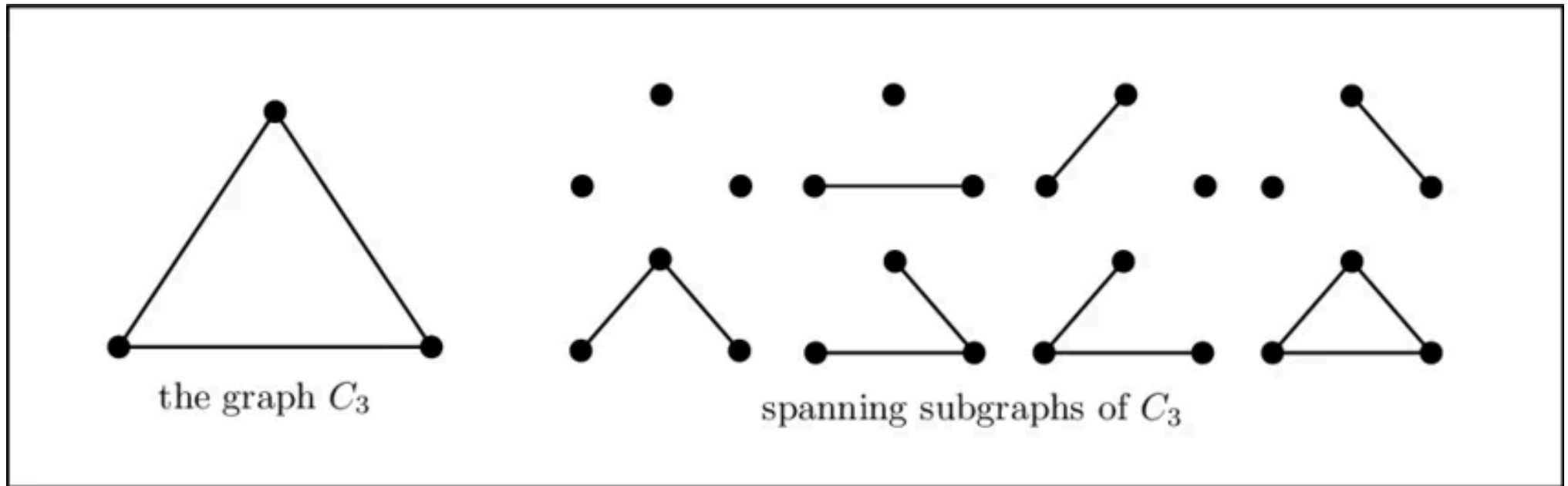


A set $M \subseteq E(G)$ of independent edges is called *matching*.

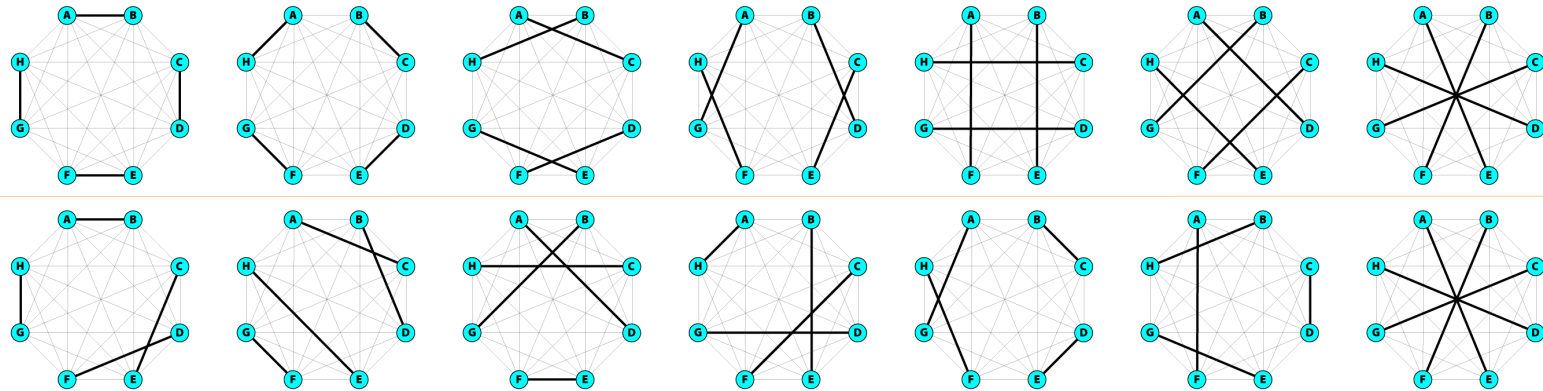


- We write $V(M)$ to denote the ends of the members of M .
- We often treat M as the subgraph of G given by $(V(M), M)$.

A subgraph $H \subseteq G$ satisfying $V(H) = V(G)$ is said to be a *spanning* subgraph of G



Definition 1 A matching $M \subseteq G$ is called *perfect matching* or a *1-factor* if M is spanning G .

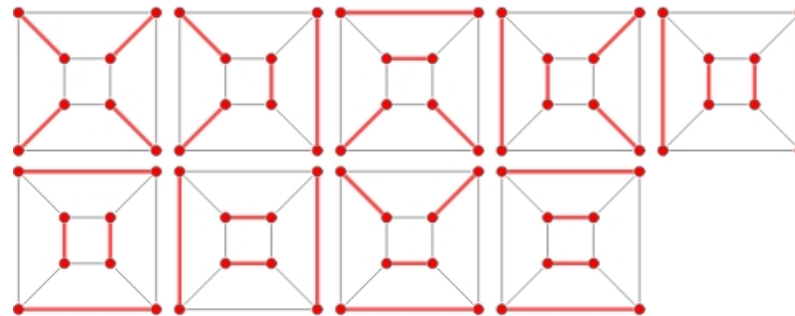


The following problem is of clear interest

Problem 2 Given a graph G , find a maximum matching in G .

The max-matching problem is in P

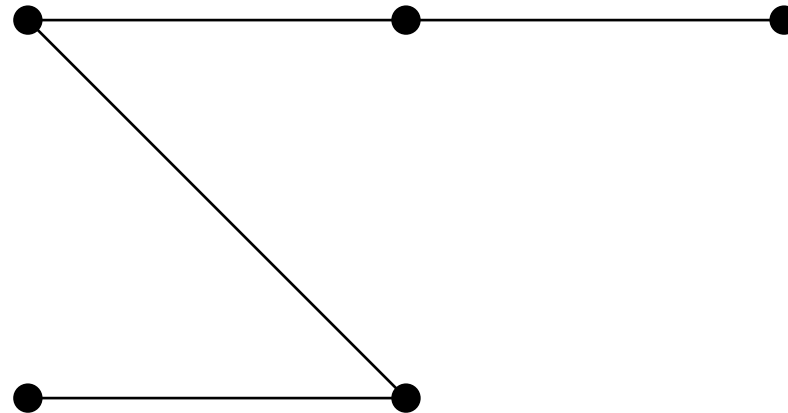
- For general graphs:
 - Edmonds “blossom” Algo.
 - Hopcroft - Karp $O(e(G)\sqrt{v(G)})$ - time.
- For bipartite graphs:
 - Via max-flow
 - Hungarian Method



2. Vertex Cover

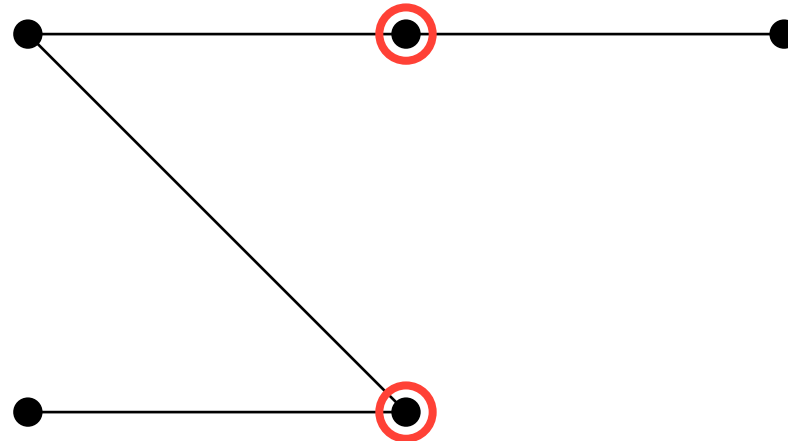
Definition 3 A set $S \subseteq V(G)$ is said to be a *vx-cover* of G if

$$\forall uv \in E(G) : u \in S \text{ or } v \in S.$$



Definition 4 A set $S \subseteq V(G)$ is said to be a *vx-cover* of G if

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- Trivially, $V(G)$ forms a *vx-cover* of G for every G .

Consequently, we are interested in the following

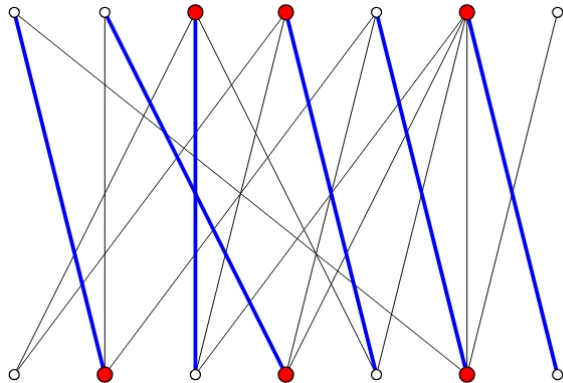
Problem 5 Given a graph G , find a minimum *vc-cover* of G .

For general graphs, the min. vx-cover problem is NPC.

Let G be a graph. denote:

- $\nu(G) :=$ size of *max* matching in G .
- $\tau(G) :=$ size of *min* vx-cover in G .

Trivially, $\tau(G) \geq \nu(G)$

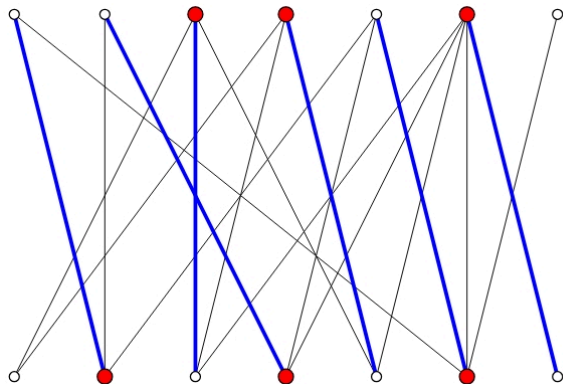


Any min-vc must “invest” at least 1 vertex for each edge in the matching.

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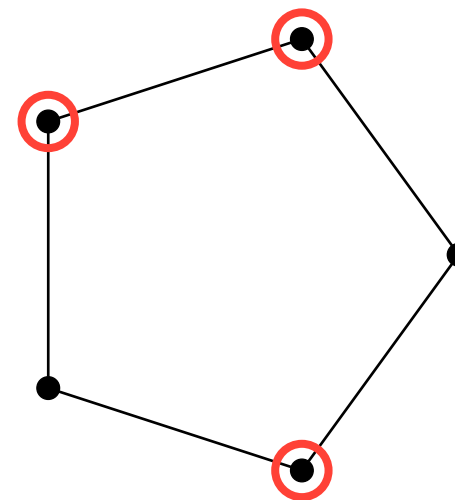
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Any min-vc must “invest” at least 1 vertex for each edge in the matching.

$\tau(G) > \nu(G)$ possible



Goal : Find a family of graphs $\mathcal{G}$ for which it holds that

$$\tau(G) = \nu(G) \quad \forall G \in \mathcal{G}$$

Then, we obtain a poly-time algo. for *min-vc* in that family.

 Remark

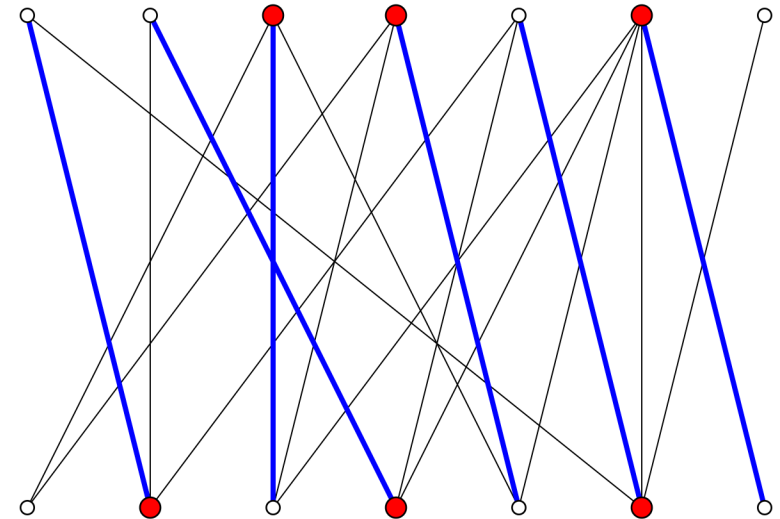
We can run any max-matching algo, and calculate $\tau(G) = \nu(G)$, thus solving the decision problem

$$k - VC := \{G : G \text{ has a vc-cover of size } k\}$$

In general we know that $\tau(G) > \nu(G)$ can happen!

Theorem 1 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

- $\nu(G) :=$ the number of edges in a maximum matching of G .
- $\tau(G) :=$ the number of vertices in a minimum vertex cover of G .



Theorem 2 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

$A \hookrightarrow B$ means
that there exists a matching
that matches all A into B .

$|N_G(S)| \geq |S|$ means that the amount of
neighbours that S has is at least as big
as than the size of S .

$$N_G(u) := \{v \in V(G) : uv \in E(G)\}$$

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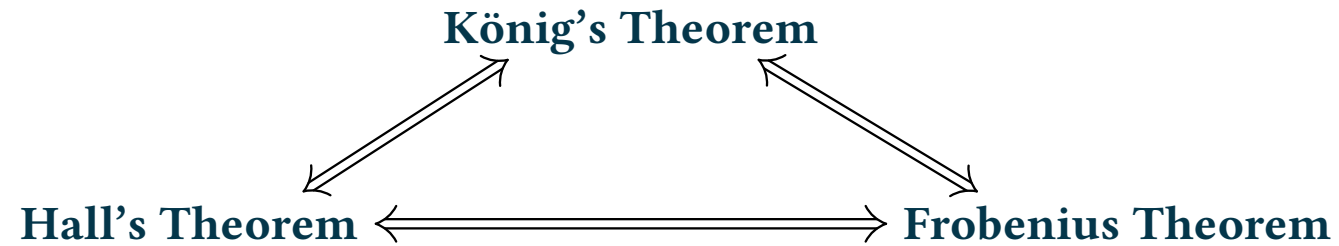
$$N_G(S) := \{v \in V(G) : \exists u \in S : v \in N_G(u)\}$$

If $|N_G(S)| \geq |S|, \forall S \subseteq A$, then we say that A satisfies the Hall-Condition!

Theorem 4 (Frobenius) Let $G := (A \cup B, E)$ be bipartite

$$\begin{array}{ccc} G \text{ has a} & & |A| = |B| \\ \text{perfect matching} & \iff & + \\ & & A \text{ or } B \text{ satisfies} \\ & & \text{the Hall-Condition} \end{array}$$

If $|N_G(S)| \geq |S|$, $\forall S \in A$, then we say that A satisfies the Hall-Condition!



We need to prove:

Theorem 5 (Frobenius) Let $G := (A \cup B, E)$ be bipartite

G has a perfect matching $\iff |A| = |B|$ and A or B satisfies the Hall-Condition

We have:

Theorem 6 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

We need to prove:

Theorem 7 (Frobenius) Let $G := (A \cup B, E)$ be bipartite

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We have:

Theorem 8 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

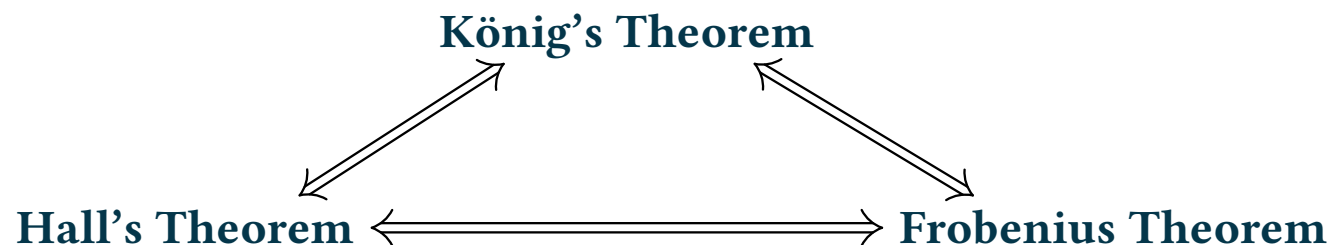
$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

$\implies$

- If G has a perfect matching then by definition $A \hookrightarrow B$
- By definition of $A \hookrightarrow B$, $|A| \leq |B|$.
- By Hall's theorem A satisfies the Hall condition
- By the same argument we can say the same thing about B

$\impliedby$

- By the Hall Theorem we have $A \hookrightarrow B$
- We have a matching M satisfying $|M| = |A|$
- since $|A| = |B|$ it follows that M is a perfect matching.



Frobenius Theorem $\Rightarrow$ König's Theorem:

Since $\tau(G) \geq \nu(G)$ is known in general, it remains to show $\tau(G) \leq \nu(G)$.

if G has a perfect matching M so that $\nu(G) = \frac{n}{2}$ and $|A| = \frac{n}{2}$. We may take A as a vertex cover of G and we are done.

Or in general we can pick 1 vertex per edges of M in a smart way.

What if G has no *perfect matching*?

hint: The idea is to “remove” unmatched vertices until we are left with a subgraph having a perfect matching. Then, think what can be done about the removed vertices.

the rest is left for the reader as an exercise

Theorem 9 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

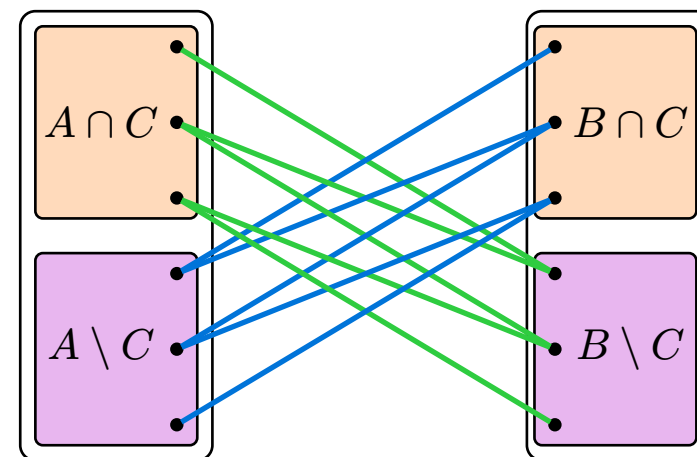
Theorem 10 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

- Let $G := (A \cup B, E)$ be bipartite, it remains to prove that $\tau(G) \leq \nu(G)$
- We seek to prove that G has a matching of size at least the minimum VC of G**
- Let $C \subseteq V(G)$ be **min-vc** of G and define

$$H := G(C \cap A, B \setminus C)$$

$$H' := G(C \cap B, A \setminus C)$$



Theorem 11 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

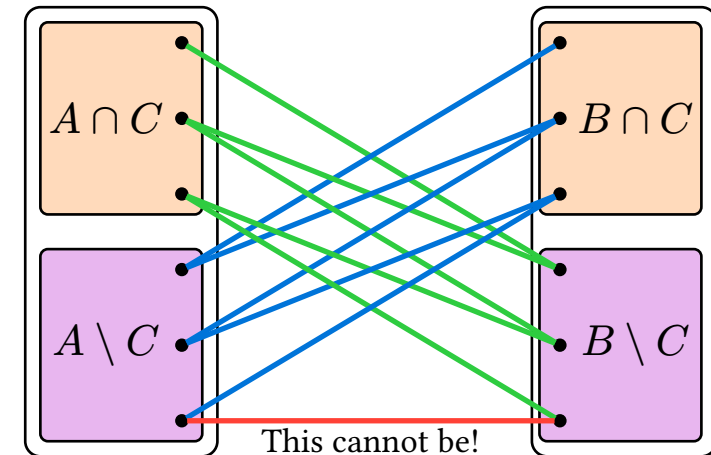
Theorem 12 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

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This cannot be!
If such an edge exists then
 C does not cover it.

Theorem 13 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

Theorem 14 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

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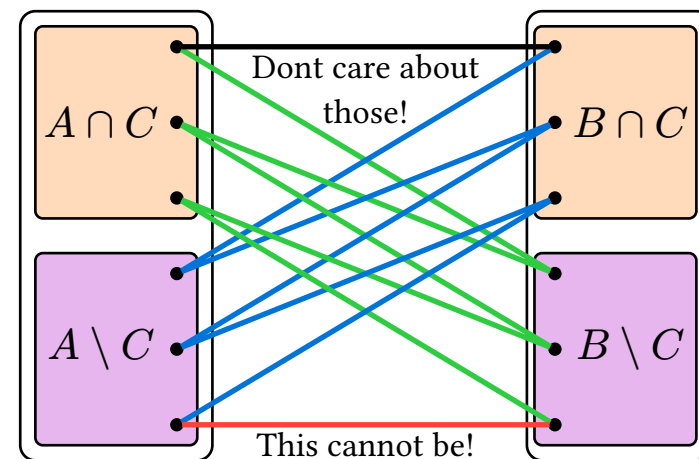
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$$H' := G(C \cap B, A \setminus C)$$

- Then, note that

$$\tau(G) = |C| = |C \cap B| + |C \cap A|$$



If such an edges exists then C doesnt cover it.

Theorem 15 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

Theorem 16 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

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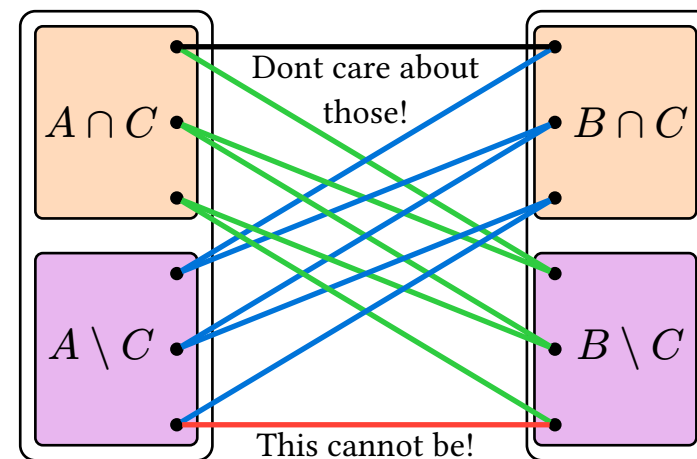
$$H' := G(C \cap B, A \setminus C)$$

- Then, note that

$$\tau(G) = |C| = |C \cap B| + |C \cap A|$$

Goal 1: Prove that

$$(C \cap A) \hookrightarrow (B \setminus C) \text{ and } (C \cap B) \hookrightarrow (A \setminus C).$$



If such an edges exists then C doesnt cover it.

Goal 1: Prove that

$$(C \cap A) \hookrightarrow (B \setminus C) \text{ and } (C \cap B) \hookrightarrow (A \setminus C).$$

 **Remark**

We prove only that $(C \cap A) \hookrightarrow (B \setminus C)$ as the other case is identical.

Assuming **Goal 1** holds, we have two matchings

$$M := (C \cap A) \hookrightarrow (B \setminus C) \text{ and } M' := (C \cap B) \hookrightarrow (A \setminus C)$$

since both are independent we conclude that $M \cup M'$ is a matching in G .

So that

$$\nu(G) \geq |M \cup M'| \geq |C \cap A| + |C \cap B| \geq |C| = \tau(G).$$

Goal 2: By Hall theorem, it remains to show

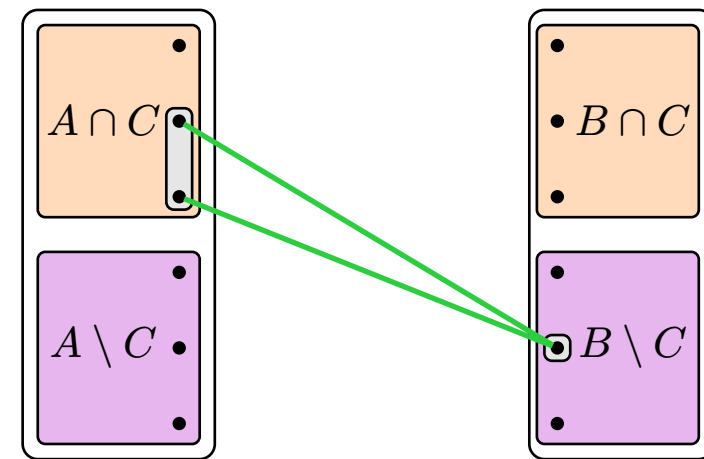
$$\forall S \subseteq (C \cap A), |N_H(S)| \geq |S|.$$

Theorem 17 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

Assume by contradiction: $\exists S \subseteq (C \cap A)$ s.t. $|N_H(S)| < |S|$.

- Set $C^* := (C \setminus S) \cup (N_H(S))$
- C^* is a vertex cover of G
- $|C^*| < |C| = \tau(G)$.
- Contradict the minimality of $\tau(G)$.



Theorem 18 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

$\implies$ Assume toward contradiction that $\exists S \subseteq A$ s.t. $|S| > |N_G(S)|$.

Then, it follows that in any matching of G at least 1 vertex of S doesn't get matched into B .

This contradicts the fact that a matching $A \hookrightarrow B$ exists.

Theorem 19 (Hall) Let $G := (A \cup B, E)$ be a bipartite graph. Then,

$$A \hookrightarrow B \iff |N_G(S)| \geq |S|, \forall S \subseteq A.$$

We may now assume that G is a graph satisfying the Hall condition, and that $\tau(G) = \nu(G)$.

Theorem 20 (König) $\tau(G) = \nu(G)$, whenever G is bipartite.

Goal : Show that $\tau(G) \geq |A|$.

Then, we have that

$$\text{size of maximum matching} = \underbrace{\nu(G) = \tau(G)}_{\text{by König}} \geq |A|.$$

Goal : Show that $\tau(G) \geq |A|$.

Proof. Let C be a minimum vertex cover in G .

$$\tau(G) = |C| = |A \cap C| + |B \cap C| \quad (1)$$

$$\bullet \quad \underbrace{N_G(A \setminus C) \subseteq B \cap C}_{\text{Otherwise } \exists uv \in E(G) \text{ not covered by } C} \implies |B \cap C| \geq |N_G(A \setminus C)|.$$

$$\bullet \quad \text{Fix } S = (A \setminus C) \quad \underbrace{\implies}_{\text{By assumption}} |N_G(A \setminus C)| \geq |A \setminus C|.$$

$$\implies |B \cap C| \geq |N_G(A \setminus C)| \geq |A \setminus C| \quad (2)$$

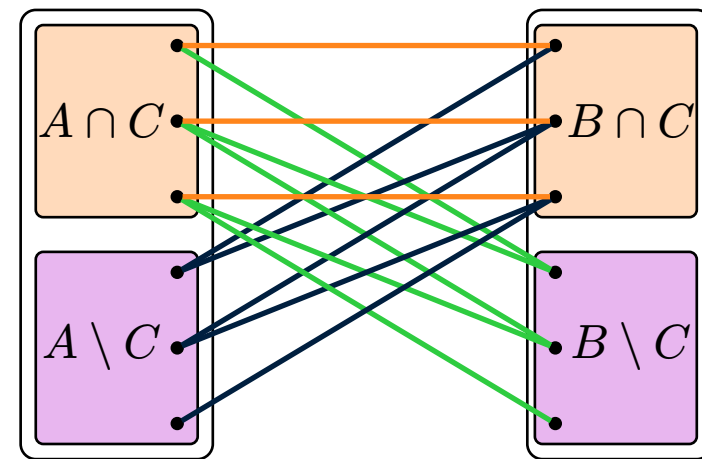
Combining Equation 1 and Equation 2

$$\tau(G) \geq |A \cap C| + |A \setminus C| = |A|$$

□

Assumption :

- $G := (A \cup B, E)$ a bipartite graph,
- $\forall S \subseteq A : |N_G(S)| \geq |S|$.



3. Berge's Theorem

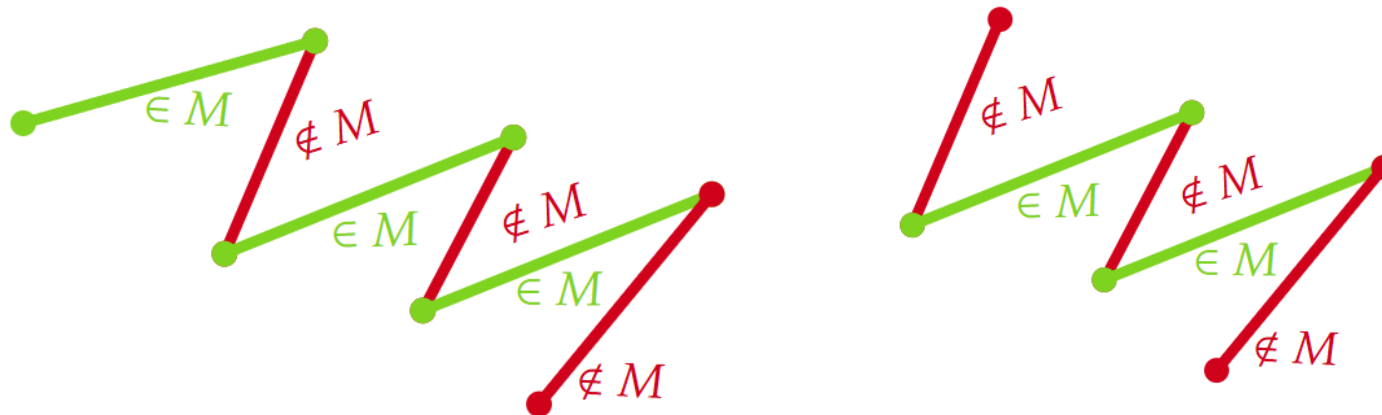
G -graph, M -matching.

A path $P := x_1x_2...x_\ell$ is called M -alternating if:

$$x_1x_2 \in M \Rightarrow x_2x_3 \notin M \Rightarrow x_3x_4 \in M \Rightarrow \dots \quad (3)$$

or

$$x_1x_2 \notin M \Rightarrow x_2x_3 \in M \Rightarrow x_3x_4 \notin M \Rightarrow \dots \quad (4)$$



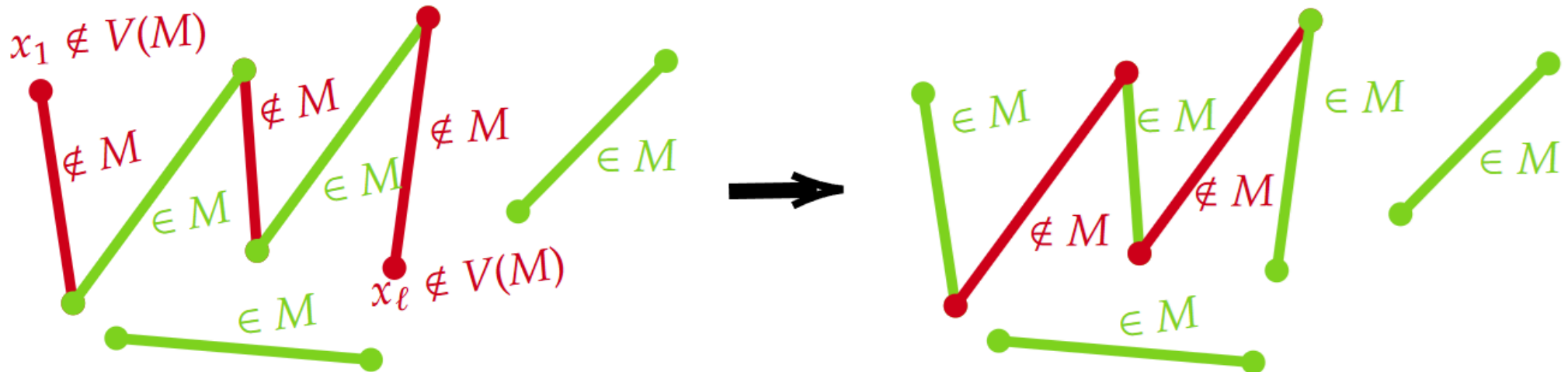
Definition 16 G -graph, M -matching.

An M -Alternating path $P := x_1 \dots x_\ell$ is called M -Augmenting if $x_1 \notin V(M)$ and $x_\ell \notin V(M)$.

Remark

Trivially, if an M -Augmenting path exists, then $|M| < \nu(G)$.

Is the opposite true?



Theorem 21 (Berge's) G -graph, M -matching

M max matching $\iff M$ has no M -Augmenting paths.

For convinence we will prove the negation

M is not max $\iff M$ has an M -Augmenting paths.

We already proved $\Leftarrow$ it remains to prove $\Rightarrow$.

Given two matching N and M in G ? What does $N \triangle M$ looks like?

$$N \triangle M := (N \setminus M) \cup (M \setminus C).$$

$\Rightarrow N \triangle M$ is a graph where every vertex degree is ≤ 2

$\Rightarrow$ Every path or cycle of $N \triangle M$ must be alternating

– **otherwise we will see path uvw with $uv \in M$ and $vw \in M$ implying that M is not a matching. (or N)**

$\Rightarrow$ Connected components of $N \triangle M$ consist of *single edges, alternating paths and even cycles*.

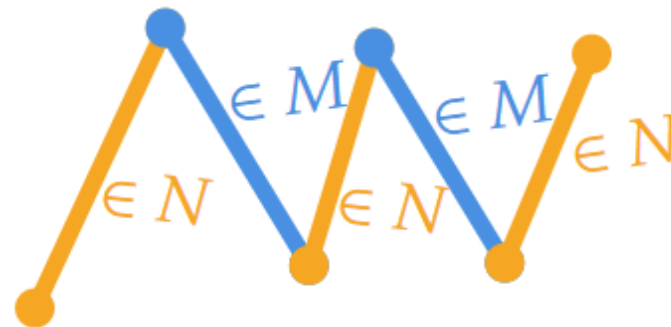


Goal : M is not max $\implies M$ has an M -Augmenting paths

Proof. Let N be a max matching in G .

- At the connected components of $N \triangle M$.
 - every connected component is either *single edge*, *alternating path* or *even cycle*.
- Since $|N| > |M|$ there must exists a connected component C of $N \triangle M$ with more edges N than edges of M .
 - C must be either a single edge of N or an M -Augmenting path. (Single edge is also M -Augmenting)

□



4. The Hungarian Method

Using the Berge's Theorem the Hungarian method for finding a max matching is the following algo:

Theorem 22 (Berge's) G -graph, M -matching
 M max matching $\iff M$ has no M -Augmenting paths.

Algorithm 1:

```

1: procedure MAX_MATCHING( $G$ )
2:   Set  $M \leftarrow \emptyset$ 
3:
4:   while there is an  $M$ -augmenting path in  $G$  do
5:     Let  $P$  be an  $M$ -augmenting path in  $G$ 
6:     Augment  $M$  along  $P$ 
7:   end
8:
9:   return  $M$ 
10: end

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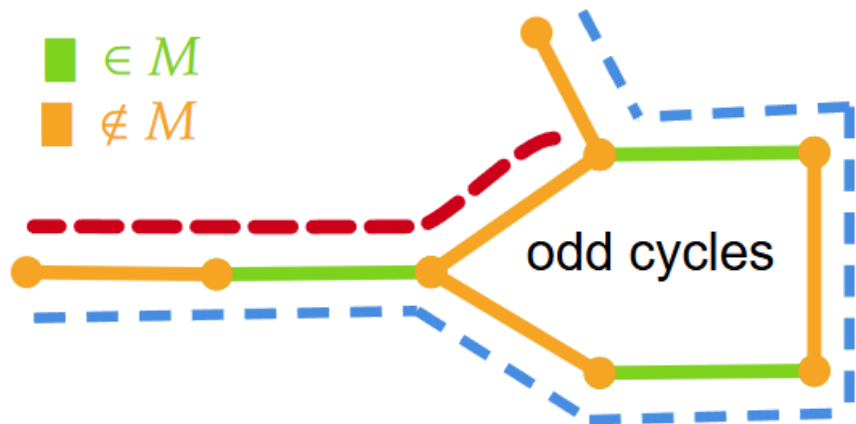
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Algorithm 2:

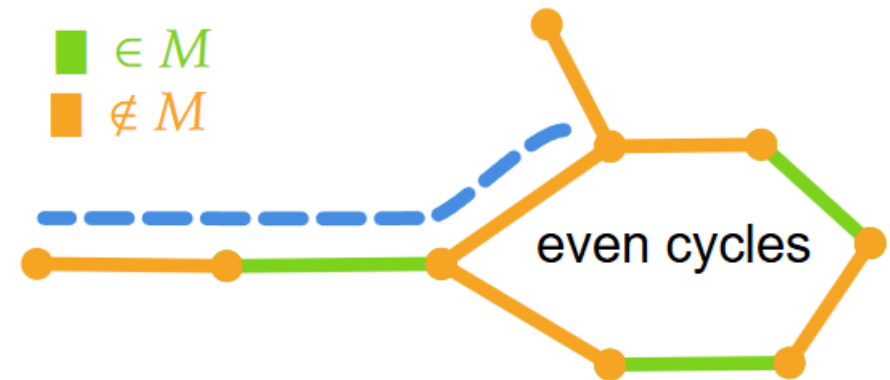
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9:   return  $M$ 
10: end
```

Problem 19 G -graph,
 M -matching.
Determine whether an
 M -augmenting path in G
exists, if it does then find
it.

In general graphs, finding an M -augmenting path might be challenging.



Going up results in a bad path,
going down results in a good one.



Here the same path is good.

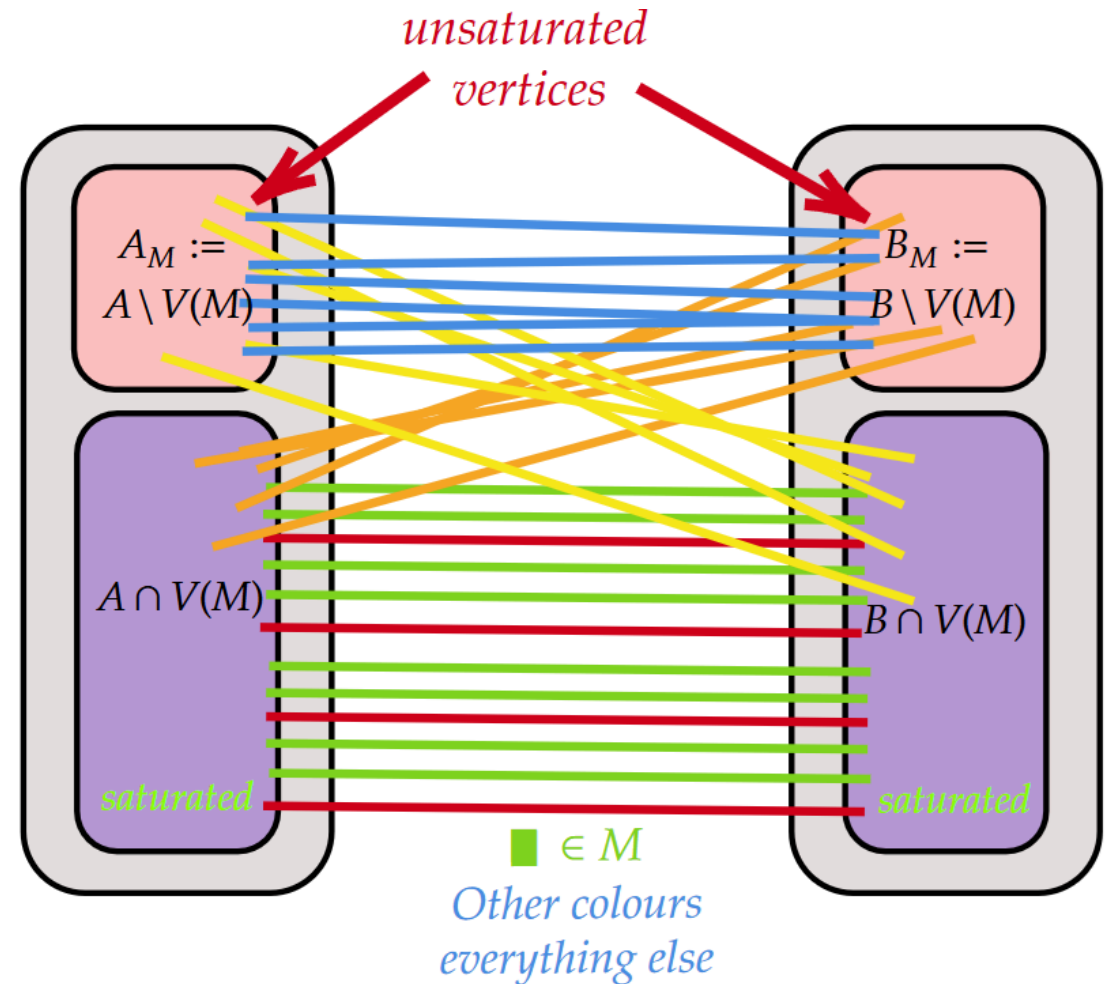
Odd cycles makes finding paths difficult.

Bipartite graphs have no odd cycles, is it easier there?

Given $G := (A \cup B, E)$ and $M \subseteq E(G)$ define

- $A_M := A \setminus V(M)$ unmatched vertices in A .
- $B_M := A \setminus V(M)$ unmatched vertices in B .

Our goal: Find an M -augmenting path from A_M to B_M . **How?**



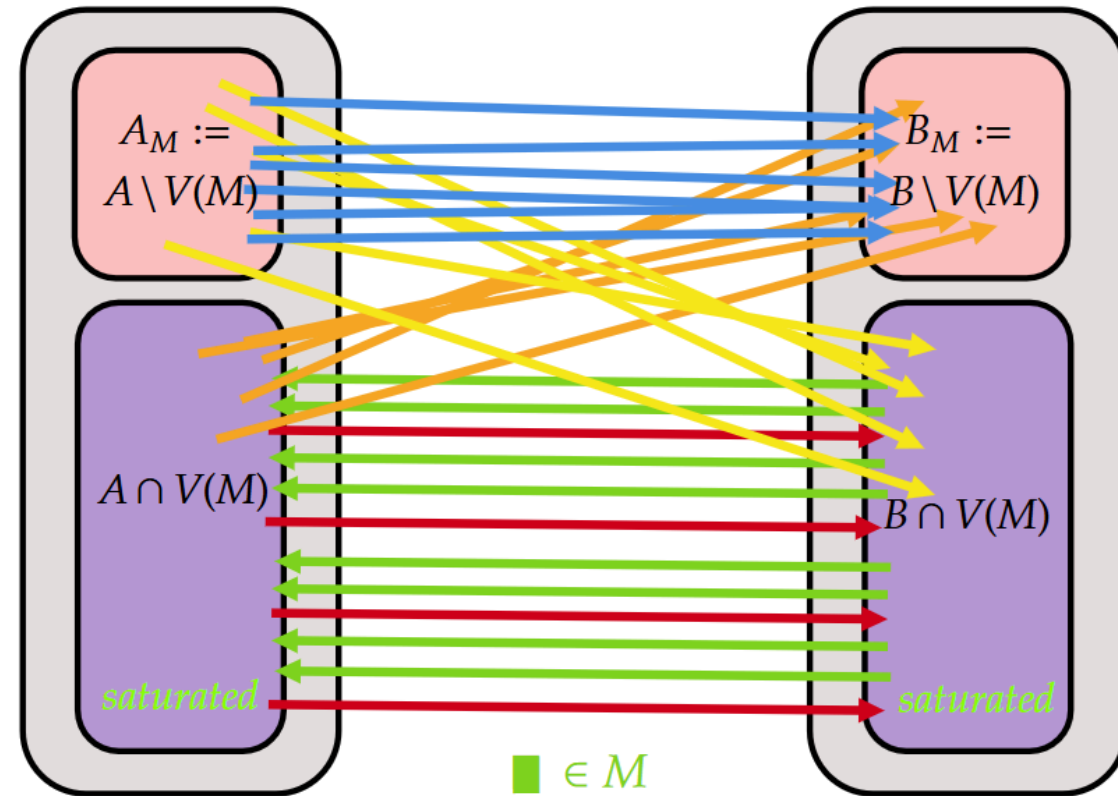
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Our goal: Find an M -augmenting path from A_M to B_M . **How?**

Construct directed graph set:

- **matching edges** M directed from B to A .
- Everything else from A to B .



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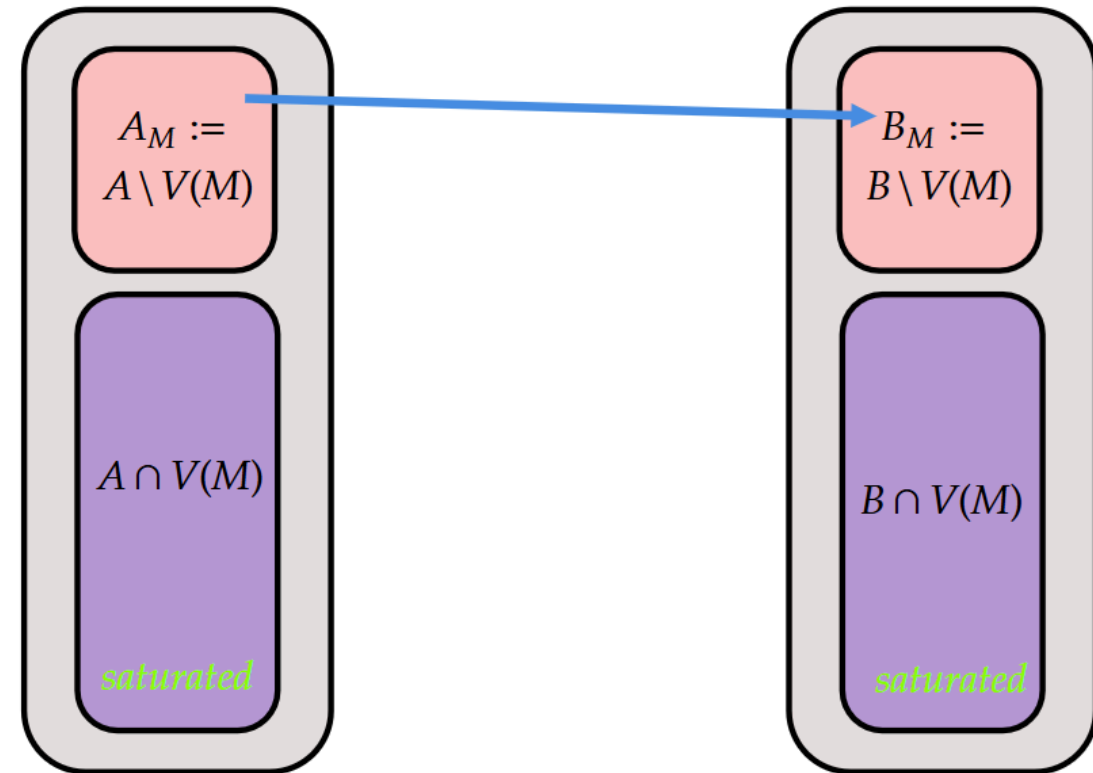
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Run any path algorithm from A_M to B_M , any such path is an M -Augmenting path.



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